

Red Hat OpenShift 5-Day Training Course

Containers · Kubernetes · OpenShift Administration

DO180 + DO280 Combined Curriculum — Revised Table of Contents

Exam	Coverage	Days
EX180	Containers & Kubernetes Fundamentals	1 – 2
EX280	OpenShift Administration	3 – 5

DAY 1

Container Fundamentals & Kubernetes Architecture

Module 1: Introduction to Containers and OpenShift

1.1 Container Concepts

- Namespaces, cgroups, Union Filesystems, OCI

1.2 Podman Basics

- podman pull / run / ps / stop / rm / exec
- Rootless containers

1.3 Container Lifecycle

- Image → Container → Running → Stopped → Removed

► [Lab: Run a UBI9 Apache container with Podman](#)

Module 2: Building Container Images

2.1 Containerfile / Dockerfile Syntax

- FROM, RUN, COPY, ADD, ENV, EXPOSE, CMD, ENTRYPOINT

2.2 Building with Podman

- podman build / tag / push

2.3 Multi-Stage Builds

- Builder stage → runtime stage pattern

2.4 Buildah for Advanced Image Construction

- buildah from / run / copy / commit
- Rootless builds, CI/CD integration

► [Lab: Build, scan, and push a multi-stage image](#)

Module 3: Kubernetes Architecture

3.1 Kubernetes High-Level Architecture

- Control Plane: kube-apiserver, etcd, kube-scheduler, kube-controller-manager
- Worker Nodes: kubelet, kube-proxy, Container Runtime (CRI-O / containerd)
- Add-ons: CoreDNS, CNI plugins, Metrics Server

3.2 Cluster Bootstrap Tools

- kubeadm init / join
- kind — Kubernetes in Docker

Note: minikube removed

3.5 Projects and Namespaces

Note: subsection bullets removed; 3.3 and 3.4 moved to Day 4

► *Lab: Bootstrap kind cluster; log in to OpenShift; explore web console*

DAY 2

Deploying & Managing Applications (DO180 / DO280)

Module 4: Projects, Namespaces, Resource Governance & Metrics

4.1 Projects and Namespaces

- Namespace-scoped vs cluster-scoped resources
- Label selectors on namespaces

4.2 ResourceQuota

- CPU, memory, object count limits per namespace
- `oc create quota` / `oc describe quota`

4.3 LimitRange

- Default / max / min resource requests per Container and Pod
- LimitRange types: Container, Pod, PersistentVolumeClaim

4.4 Metrics & Resource Monitoring

- Metrics Server — `kubectl top nodes` / `pods`
- OpenShift Observe tab — per-project Prometheus dashboard

► *Lab: Create project; apply ResourceQuota + LimitRange; observe with kubectl top*

Module 5: Core Kubernetes Workload Objects

5.1 Pod

- Smallest deployable unit; containers, initContainers, volumes, restartPolicy
- Pod phases: Pending → Running → Succeeded / Failed / Unknown

5.2 ReplicationController (legacy)

- Ensures N replicas; equality-based selector only

5.3 ReplicaSet

- Set-based label selectors; owns Pods via ownerReferences

5.4 Deployment

- RollingUpdate vs Recreate strategy
- `kubectl rollout status` / `history` / `undo`
- DeploymentConfig vs Deployment

► *Lab: Create Pod; scale with ReplicaSet; wrap in Deployment; rolling update and rollback*

Module 6: Developer Sandbox & Application Deployment

6.1 OpenShift Developer Sandbox

- Free 30-day shared cluster at developer.redhat.com/developer-sandbox
- Pre-configured namespaces, quotas, OpenShift Pipelines

6.2 Deployment Strategies

- RollingUpdate, Recreate, Blue/Green, Canary

Note: 6.3 Using `oc new-app` moved to Day 4

► *Lab: Deploy multi-tier app; configure rolling update*

DAY 3

Module 7: Advanced Workload Controllers

7.1 Horizontal Pod Autoscaler (HPA)

- CPU, memory, or custom metrics; minReplicas / maxReplicas
- kubectl autoscale; HPA v2 multiple metrics

7.2 StatefulSet

- Ordered stable Pod identities: pod-0, pod-1, pod-2
- Headless Service (clusterIP: None); volumeClaimTemplates
- Use cases: databases, Kafka, Zookeeper

7.3 DaemonSet

- One Pod per node; Fluentd, Node Exporter, CNI agents
- RollingUpdate strategy; tolerations for control-plane nodes

► [Lab: HPA on CPU-stressed app; StatefulSet with PVCs; DaemonSet on all nodes](#)

Module 8: ConfigMaps and Secrets

8.1 ConfigMaps

- --from-literal, --from-file, --from-env-file
- Consume as env vars (envFrom / valueFrom) or volume mounts

8.2 Secrets

- Types: Opaque, kubernetes.io/tls, kubernetes.io/dockerconfigjson
- oc create secret generic / tls / docker-registry
- Inject as env vars or volume mounts

Note: 8.3 Best Practices removed; module lab removed

Module 9: Persistent Storage

9.1 Storage Concepts

- PersistentVolume (PV), PersistentVolumeClaim (PVC), StorageClass

9.2 Creating PersistentVolumeClaims

- accessModes, resources.requests.storage, storageClassName
- Binding: Pending → Bound

9.3 Access Modes

- RWO — ReadWriteOnce
- ROX — ReadOnlyMany
- RWX — ReadWriteMany
- RWOP — ReadWriteOncePod

9.4 Using PVCs in Deployments

- volumes: persistentVolumeClaim + volumeMounts in container spec

9.5 StorageClasses and Dynamic Provisioning

- Default StorageClass; WaitForFirstConsumer; volume expansion

Note: 9.6 StatefulSets and Storage removed; module lab removed

DAY 4

Services, Routes, Networking, S2I & CRC (DO280)

Module 10: Services, Routes & Networking Basics

10.1 Kubernetes Services

- ClusterIP — stable virtual IP inside the cluster

10.2 NodePort

- Expose on node IP, static port 30000–32767

10.3 LoadBalancer

- Cloud provider LB integration; MetalLB for on-premises

10.4 OpenShift Routes

Note: subsection bullets removed

Note: 10.5 ConfigMaps and Secrets in Services removed

➤ *Lab: Expose app via ClusterIP; NodePort; create Edge TLS Route; test passthrough*

Module 11: Source-to-Image (S2I) & Image Streams

11.1 Source-to-Image (S2I)

- Automated build from source repo → image without Containerfile
- Builder images: Python, Node.js, Java, PHP, Ruby, .NET

11.2 OpenShift Builds & BuildConfig

- Build strategies: Source (S2I), Docker, Custom
- oc start-build; image change triggers; webhook triggers

11.3 ImageStreams

- Tracks image versions; ImageStreamTag
- Image change trigger auto-rolls out Deployment on new push
- oc import-image

➤ *Lab: Deploy Node.js via S2I BuildConfig; GitHub webhook; verify auto-rebuild*

Module 12: Cluster Networking Deep Dive

12.1 OpenShift Networking Architecture (OVN-Kubernetes)

12.2 Network Policies

12.3 Managing Network Policies

Note: subsection bullets removed from 12.1–12.3; 12.4 Egress IP and Egress Firewall removed

➤ *Lab: Deny-all NetworkPolicy; open specific ingress; verify isolation*

Module 13: CodeReady Containers (CRC) — Local OpenShift

13.1 What is CRC?

- Full OpenShift 4.x on a single developer VM (libvirt / HyperKit / Hyper-V)
- Min: 4 vCPU, 9 GB RAM, 35 GB disk

13.2 CRC Setup & Workflow

- crc setup / start / console --credentials / stop / delete

13.3 Use Cases

- Offline development; exam practice; test Operators and SCCs locally

➤ *Lab: Install CRC; deploy app; expose via Route; verify TLS termination*

Moved to Day 4

3.3 OpenShift Architecture & OpenShift vs Kubernetes

- OpenShift = Kubernetes + RHCOS + Operators + Enterprise security
- Integrated OAuth, image registry, Route controller, OLM

3.4 Connecting to OpenShift (oc login)

- oc login, kubeconfig, contexts

- Web Console: Administrator vs Developer perspective

6.3 Using oc new-app

- Auto-detects language / framework from Git repo
- Creates Deployment, Service, and ImageStream automatically

DAY 5

OpenShift Administration: Auth, RBAC, Operators & Maintenance (DO280)

Module 14: Authentication and Identity Providers

14.1 OpenShift OAuth

- Built-in OAuth server; kubeadmin bootstrap user

14.2 Configuring HTPasswd Authentication

- htpasswd file → Secret → OAuth CR identity provider

Note: 14.3 Other Identity Providers (LDAP, GitHub) removed

► *Lab: Configure HTPasswd IDP; create users; test login; remove kubeadmin*

Module 15: Role-Based Access Control (RBAC)

15.1 RBAC Concepts

- apiGroups + resources + verbs
- Role vs ClusterRole; RoleBinding vs ClusterRoleBinding

15.2 Managing Role Bindings

- oc adm policy add-role-to-user / add-cluster-role-to-user
- Built-in roles: cluster-admin, admin, edit, view

15.3 Creating Custom Roles

- Least-privilege Role / ClusterRole YAML
- Aggregated ClusterRoles; oc auth can-i --as=

15.4 Service Accounts

Note: subsection bullets removed

► *Lab: Custom Role for dev team; bind to group; verify with oc auth can-i; SA for app*

Module 16: Security Context Constraints (SCCs)

16.1 What Are SCCs?

- Built-in SCCs: restricted-v2, baseline, privileged, anyuid, hostnetwork
- Pod Security Admission (PSA): enforce / warn / audit labels

16.2 Assigning SCCs

- oc adm policy add-scc-to-user -z -n
- SCC priority; oc get pod -o yaml | grep scc

Note: 16.3 Writing SCC-Compliant Containers removed; 16.4 Pod Security Admission (PSA) removed

► *Lab: Privileged app blocked by restricted-v2; fix with custom SA + SCC*

Module 17: Operators and the Operator Lifecycle Manager (OLM)

17.1 What Are Operators?

- Controllers encoding operational knowledge via CRDs
- Capability levels: Basic Install → Auto Pilot

17.2 Operator Lifecycle Manager (OLM)

- CatalogSource, Subscription, InstallPlan, CSV, OperatorGroup

- OperatorHub tiers: Red Hat, Certified, Community

17.3 Using an Operator — Example: Cert-Manager

- Install from OperatorHub; ClusterIssuer CR; Certificate CR → TLS Secret

17.4 Helm on OpenShift

- helm install / upgrade / rollback
- Helm charts vs Operators

► [Lab: Install Cert-Manager from OperatorHub; issue TLS cert; wire into Route](#)

Module 18: Monitoring, Troubleshooting & Cluster Maintenance

18.1 OpenShift Monitoring Stack

- Prometheus, Grafana

Note: Alertmanager, Thanos, and User Workload Monitoring removed

18.3 Troubleshooting Framework

- oc get events; oc describe pod / node
- oc logs --previous; oc debug node/
- oc get co — Cluster Operators health

18.4 Common Issues and Solutions

- CrashLoopBackOff, ImagePullBackOff, Pending Pod, OOMKilled

18.5 Node Maintenance (Cordon, Drain, Uncordon)

- oc adm cordon / drain / uncordon
- Taints and Tolerations

18.6 etcd Backup

- etcd backup: cluster-backup.sh

Note: 18.2 Custom Alerting Rules, Cluster Updates portion of 18.6, and 18.7 Resource Quotas at Scale removed

► [Lab: Diagnose CrashLoopBackOff; drain node; etcd backup](#)

Day-by-Day Key Topics & Exam Mapping

Day	Key Topics	Exam
1	Containers, Podman, Buildah, K8s Architecture, kubeadm, kind, Projects	EX180
2	ResourceQuota, LimitRange, Metrics, Pod/RC/RS/Deployment, Dev Sandbox	EX180
3	HPA, StatefulSet, DaemonSet, ConfigMap, Secret, PV/PVC/StorageClass	EX180/EX280
4	Services (NodePort/LB), Routes, S2I, BuildConfig, ImageStreams, NetworkPolicy, CRC, oc login, oc new-app, OpenShift Architecture	EX280
5	Auth/IDP, RBAC, SCCs, Operators/OLM/Helm, Prometheus/Grafana, Troubleshooting, Node Drain, etcd Backup	EX280

All section, module, and lab edits above were applied directly against the uploaded course content; numbering follows this document's own structure.